# OMITTING TYPES FOR INFINITARY REAL-VALUED LOGIC: A TOPOLOGICAL PERSPECTIVE

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ABSTRACT. We extend Keisler's omitting types theorem for countable fragments of the infinitary logic with countable conjunctions and disjunctions to the corresponding extension of his model theory for real-valued structures. In the passage from two-valued to real-valued logic, equality becomes a uniform structure, a metric for countable vocabularies, and the axioms of equality become axioms of continuity. Keisler's general structures, whose predicates are arbitrary bounded functions, are therefore the real-valued form of first-order structures without equality, as the metric structures of Eagle's theorem are the real-valued form of structures with equality. Accordingly, we build models from closed terms, as Keisler and Miller do in logic without equality, and we show that the space of Henkin types of a countable fragment is Čech-complete. By Morley's Baire category method, each theory on the fragment with a model then has one with countable sorts omitting any countably many types with empty interior in the subspace of Henkin types satisfying the theory.

## 1. INTRODUCTION

We prove the omitting types theorem for countable fragments of the infinitary $L_{\omega_1\omega}$ extension of Keisler's model theory for real-valued structures [Kei23], and the proof brings together four works of Keisler. His book on $L_{\omega_1\omega}$ [Kei71, Chapter 11] contains the theorem for countable fragments of classical $L_{\omega_1\omega}$, which is the statement extended here. His article on forcing [Kei73] contains a general omitting types theorem, whose hypothesis reappears in topological form in Theorem B below. Keisler and Miller [KM01, Section 4] determined the form that the theorem takes in first-order logic without equality, in which the model is built from closed terms and the types to be omitted are tested through tuples of terms. Keisler's general structures [Kei23, Section 2.1], finally, interpret the function and predicate symbols by arbitrary mappings, the predicates being bounded, and they are the structures of our logic.

The four works belong to two lines of development, that of the omitting types theorem and that of equality. In 1969, Makkai extended Smullyan's method of consistency properties to the infinitary logics on admissible sets [Mak69]. In 1971, Keisler made the model existence theorem for consistency properties a central method of his book on $L_{\omega_1\omega}$ [Kei71, Chapter 3]. In 1973, Keisler built countable models by forcing with finite sets of sentences satisfiable in a given class of models [Kei73, p. 100, Example 1.2]. In the same work, he replaced principal types by $\forall\exists$ formulas over a set of formulas of a countable fragment [Kei73, p. 107, Definition]. His general omitting types theorem gives conditions under

which a $\forall\exists$ class contains a countable model in which countably many given $\forall\exists$ sentences hold [Kei73, p. 108, General Omitting Types Theorem]. In 1974, Morley derived the omitting types theorem for countable fragments from the Baire category theorem, applied to a space of countable structures, which he showed to be a Baire space [Mor74]. In 2001, Keisler and Miller carried the theorem into first-order logic without equality, in which principal types give way to locally realized types [KM01, Section 4]. In 2014, Eagle applied Čech's form of the Baire category theorem to a space of countable metric structures, drawing in part on work of Caicedo and Iovino [Eag14, CI14]. The predicates of these structures take values in $[0, 1]$ and are uniformly continuous with respect to a master metric [Eag14, Definition 1.4]. The moduli of uniform continuity are prescribed by the signature [Eag14, Definition 1.5]. The master metric is the real-valued counterpart of equality in two precise senses: two elements are equal exactly when their distance is zero, and the uniform continuity of every predicate and function symbol with respect to it is the real-valued counterpart of the substitutivity of equals in atomic formulas.

The second line begins with the axioms of equality in first-order logic, which assert its reflexivity and the interchangeability of equal elements in every atomic formula. These axioms make equality a congruence for the function and predicate symbols. In real-valued logic, their counterparts are axioms of continuity.[1] A general form of these axioms, uniform continuity with moduli prescribed by the signature, was the setting of Iovino's thesis of 1994 on stable Banach space structures [Iov94]. In this setting, the role of equality passes to a uniform structure [Iov94]. For a countable signature, the uniform structure has a countable base, and it is therefore induced by a single pseudometric (see [Eng89, Section 8.1]). Iovino's papers of 1999 on stable Banach spaces rest on the same framework [Iov99a, Iov99b], and in 2002 Henson and Iovino presented the model theory of normed space structures in it [HI02, Sections 2 and 3]. In 2010, Ben Yaacov and Usvyatsov built moduli of uniform continuity into the signatures of continuous logic [BU10, Section 2]. In 2023, Keisler obtained the pseudometric from the formulas themselves, for vocabularies with countably many function and predicate symbols [Kei23, Section 3.3]. Every general structure for such a vocabulary expands to one with a distance predicate that is a uniform limit of formulas [Kei23, Theorem 3.3.4]. This distance predicate is unique up to uniform equivalence, and every formula is uniformly continuous with respect to it [Kei23, Section 3.2 and Proposition 3.3.7].

The two lines meet in Keisler's comparison of structures: general structures correspond to first-order structures without equality, as metric structures with a distance predicate correspond to first-order structures with equality [Kei23, Section 1]. The passage from Eagle's structures to Keisler's is therefore the real-valued counterpart of the passage from logic with equality to logic without equality. For

[1] In Łukasiewicz predicate logic, the congruence axioms for a similarity, or graded equality, make every predicate 1-Lipschitz for the distance given by one minus the similarity (see [Háj98, Section 5.6] and the earlier work collected there).

the second passage the analysis of Keisler and Miller is the precedent, and we follow it in building our models from closed terms.

In the infinitary extension of Keisler's model theory, each sentence takes its values in a bounded interval of its own. Hence the formal types, families of such values, are the points of a product of compact intervals that vary from sentence to sentence. We expand the vocabulary by countably many new constants of each sort. Since the vocabulary has no equality symbol, the closed terms of the expanded vocabulary serve themselves as the elements of a structure, with no quotient to be taken. The Henkin conditions are the real-valued form of Henkin's requirement of witnesses. The Henkin types, defined on the expanded language by these conditions, are exactly the types of such term structures. The Henkin types of a fragment are defined likewise on the fragment that it generates in the expanded language. For a countable vocabulary and a countable fragment, we show that the space of these Henkin types, with the product topology, is Čech-complete. Morley's method therefore operates in this space, each point of which is the type of a term structure.

A type of the fragment in a finite tuple $\bar{x}$ of variables is a set of expressions $\varphi(\bar{x}) \geq r$, with $r$ rational and with $\varphi$ a formula of the fragment whose free variables are among $\bar{x}$. The type is omitted by a Henkin type of the fragment that satisfies none of the instances obtained by substituting closed terms of the expanded vocabulary for these variables.

Keisler and Miller call a type locally realized when it is implied, through some tuple of terms, by a formula consistent with the theory [KM01, Definition 4.1]. Every type realized in every model of a consistent theory is locally realized [KM01, Theorem 4.2]. Caicedo and Iovino use tuples of terms in the same way in their $\omega$-principal types [CI14, Definition 3.5]. Eagle does so in his principal types, and he credits this use of terms to Keisler and Miller and to Caicedo and Iovino [Eag14, Definition 4.10]. Keisler and Miller also state local realization topologically, as the existence of a term mapping between spaces of types under which the preimage of the type has nonempty interior [KM01, Section 4]. The hypothesis of Theorem B is a condition of the opposite kind, stated in the space of Henkin types that satisfy the theory. For every tuple of closed terms, the members of this space that satisfy the instance of the type at those terms are required to form a set with empty interior. The hypothesis of Keisler's general omitting types theorem can be read in the same way [Kei73, p. 108, General Omitting Types Theorem]. We consider the case in which his $\forall\exists$ formulas are taken over the set of all formulas of the fragment. A model omits a type $\Gamma(\bar{x})$ of the fragment exactly when it satisfies the sentence $\forall\bar{x} \bigvee_{\gamma\in\Gamma} \neg\gamma(\bar{x})$, which has the $\forall\exists$ form. For this sentence, Keisler asks that every finite set of sentences satisfiable in the class remain satisfiable after the addition of $\bigvee_{\gamma\in\Gamma} \neg\gamma(\bar{c})$, for every tuple $\bar{c}$ of new constants. Equivalently, no finite set of sentences satisfiable in the class defines a subclass that consists of models of the instance $\Gamma(\bar{c})$. In topological terms, every instance has empty interior with respect to the basic sets given by finite sets of sentences. In this case, moreover, every $\forall\exists$ sentence $\forall\bar{x} \bigvee_i \theta_i(\bar{x})$ over the fragment holds in exactly the models that omit the type $\{\neg\theta_i\}_i$; hence this reading covers every sentence to which the theorem applies. Theorem B differs from Keisler's theorem in three respects. Its space is

the space of Henkin types that satisfy the theory, and its basic open sets are given by finite theories. Its instances are taken at tuples of closed terms, as in the work of Keisler and Miller, rather than at tuples of constants. Its types consist of closed conditions $\varphi \geq r$ in place of formulas. With these differences, Theorem B extends this case of the general omitting types theorem to Keisler's general structures. It also explicates the hypothesis of that theorem as a statement about interiors in a Čech-complete space.

**Theorem A.** Let a countable vocabulary be given, together with a formal type for a set of sentences of $L_{\omega_1\omega}$ for this vocabulary. The formal type is realized by a structure whose sorts are countable if and only if it extends to a Henkin type. If the given sentences belong to a countable fragment, then the formal type is realized by some structure if and only if it extends to a Henkin type of the fragment.

**Theorem B.** Let a countable fragment of $L_{\omega_1\omega}$ for a countable vocabulary be given, together with a theory on sentences of the fragment that has a model and a sequence of types of the fragment. Suppose that, for every instance of every type of the sequence, the Henkin types of the fragment that satisfy the theory and the instance form a set with empty interior in the space of Henkin types of the fragment that satisfy the theory. Then some Henkin type of the fragment that satisfies the theory omits every type of the sequence; consequently, the theory has a model whose sorts are countable and in which no tuple of elements satisfies any type of the sequence.

**Organization of the paper.** In Section 2 we recall Čech's form of the Baire category theorem; in Sections 3–8 we set up the structures, languages, fragments, theories and formal types. In Section 9 we define Henkin types and prove Theorem A. In Section 10 we show that, in the countable case, the space of Henkin types of a fragment is a $G_\delta$ subset of the product of intervals in which it lies. Theorem B, proved in Section 11, then follows from Čech's form of the Baire category theorem.

## 2. Topological Preliminaries

The topological fact on which the paper rests is Čech's form of the Baire category theorem. The class of spaces to which Čech's theorem applies contains every countable intersection of open subsets of a compact Hausdorff space. The argument of the paper takes place in the space of Henkin types for the sentences of a fragment of the infinitary language (see Section 9). In Section 10 we show that this space of Henkin types is such an intersection when the vocabulary and the fragment are countable. In this section we recall the definitions that Čech's theorem involves, with Engelking's *General Topology* [Eng89] as our reference for topology.

A topological space is a **Baire space** if the intersection of every sequence of dense open subsets of the space is dense; Baire spaces were introduced by Bourbaki in 1948 (see [Eng89, notes to Section 3.9]). The spaces to which this property

will be applied are Tychonov spaces, that is, completely regular $T_1$-spaces; every subspace of a compact Hausdorff space is one (see [Eng89, Theorem 3.2.6]). Let $X$ be a Tychonov space, and let $(\mathscr{U}_n)_{n<\omega}$ be a sequence of open covers of $X$. Completeness of this sequence concerns the families $(F_i)_{i\in I}$ of closed subsets of $X$ that have the finite intersection property and, for every $n < \omega$, a member contained in some element of $\mathscr{U}_n$. The sequence $(\mathscr{U}_n)_{n<\omega}$ is **complete** if every such family has nonempty intersection. The space $X$ is **Čech-complete** if it has a complete sequence of open covers. Čech [Čec37] introduced these spaces in 1937, under the name of topologically complete spaces, as the Tychonov spaces that are $G_\delta$ subsets of their Stone–Čech compactifications. The characterization of Čech-complete spaces by sequences of open covers, adopted here as the definition, was established independently by Frolík in 1960 and by Arhangel'skiĭ in 1961 (see [Eng89, Theorem 3.9.2 and the notes to Section 3.9]). By Theorems 3.9.1 and 3.9.2 of Engelking [Eng89], a Tychonov space is Čech-complete if and only if it is a $G_\delta$ subset of some, or equivalently of every, compactification of it. Consequently, every $G_\delta$ subset $X$ of a compact Hausdorff space $K$ is Čech-complete. Indeed, the closure of $X$ in $K$ is compact (see [Eng89, Theorem 3.1.2]), and is therefore a compactification of $X$ in which $X$ is a $G_\delta$ set. The compact Hausdorff spaces in which such $G_\delta$ subsets will lie are products $\prod_{i\in I}[a_i, b_i]$, endowed with the product topology, of intervals indexed by a set $I$, with real endpoints $a_i \leq b_i$. Each interval $[a_i, b_i]$ is compact (see [Eng89, Theorem 3.2.8]), and so is the product, by Tychonov's theorem (see [Eng89, Theorem 3.2.4]). Čech also proved the following form of the Baire category theorem (see [Eng89, Corollary 3.9.4 and the notes to Section 3.9]).

**Theorem 1** (Baire category theorem; Čech [Čec37])**.** Every Čech-complete space is a Baire space.

## 3. Vocabularies and Structures

The models whose existence the omitting types theorem asserts are real-valued structures. Such a structure consists of one or more nonempty sets, called its sorts, of functions from finite products of sorts to sorts, and of its predicates. The predicates of the structure are bounded real-valued functions on finite products of sorts. Classical first-order structures for vocabularies without equality correspond to the case in which every predicate takes only the values $1$ and $-1$, read as true and false, respectively.

Formulas speak about structures through a vocabulary, which is a set of symbols for sorts, functions and predicates. A structure for the vocabulary interprets each symbol as an actual sort, function or predicate. Since symbols must be kept apart from their interpretations, the symbols of a vocabulary are printed in this color, and mathematical objects in black. For the same reason, the variables, the connectives and the formal expressions built from them are printed in this color.

A **sort descriptor** is a purely syntactic symbol $M$. Let $\mathbb{M}$ be a nonempty set of sort descriptors. A **function symbol for** $\mathbb{M}$ is a purely syntactic symbol $f$ given together with

(i) a natural number $n = n_f$, called the **arity of** $f$, and
(ii) an $(n+1)$-tuple $(M_i)_{i\leq n}$ of sort descriptors in $\mathbb{M}$, called the **extended arity of** $f$.

We indicate the arity and the extended arity of $f$ by writing

$$f\colon \prod_{i<n} M_i \to M_n.$$

A function symbol of arity 0 is called a **constant symbol of sort** $M_0$. A **predicate symbol for** $\mathbb{M}$ is a purely syntactic symbol $p$ given together with

(i) a natural number $n = n_p$, called the **arity of** $p$,
(ii) an $n$-tuple $(M_i)_{i<n}$ of sort descriptors in $\mathbb{M}$, called the **extended arity of** $p$, and
(iii) a real number $r = r_p \geq 0$, called the **a priori bound of** $p$.

We indicate the arity, the extended arity and the a priori bound of $p$ by writing

$$p\colon \prod_{i<n} M_i \to [-r, r].$$

A **vocabulary** is a triple $V = (\mathbb{M}, F, P)$ consisting of a nonempty set $\mathbb{M}$ of sort descriptors, a set $F$ of function symbols for $\mathbb{M}$, and a set $P$ of predicate symbols for $\mathbb{M}$. The set $P$ is required to contain a nullary predicate symbol $1$, called the **logical unity constant**, with a priori bound $r_1 = 1$. The vocabulary $V$ is **countable** if the sets $\mathbb{M}$, $F$ and $P$ are countable. The a priori bounds provide the compactness on which the topology of the paper rests. Indeed, as we show in Section 4, the value of a formula in a structure lies in an interval $[-r, r]$ determined by the formula alone. Consequently, the types of all structures lie in a product of compact intervals, the compact Hausdorff space of formal types of Section 8.

Given a vocabulary $V = (\mathbb{M}, F, P)$, a $V$**-structure**, or real-valued structure with vocabulary $V$, is a triple

$$\mathfrak{M} = \langle (M^{\mathfrak{M}})_{M\in\mathbb{M}}, (f^{\mathfrak{M}})_{f\in F}, (p^{\mathfrak{M}})_{p\in P}\rangle$$

in which

- the interpretation $M^{\mathfrak{M}}$ of each sort descriptor $M \in \mathbb{M}$ is a nonempty set,
- the interpretation of each function symbol $f\colon \prod_{i<n} M_i \to M_n$ in $F$ is a function
$$f^{\mathfrak{M}}\colon \prod_{i<n} M_i^{\mathfrak{M}} \to M_n^{\mathfrak{M}},$$
- the interpretation of each predicate symbol $p\colon \prod_{i<n} M_i \to [-r_p, r_p]$ in $P$ is a function
$$p^{\mathfrak{M}}\colon \prod_{i<n} M_i^{\mathfrak{M}} \to [-r_p, r_p],$$
and
- the interpretation of the logical unity constant is $1^{\mathfrak{M}} = 1$.

Here a nullary symbol is interpreted on the one-point set $\prod_{i<0} M_i^{\mathfrak{M}}$, and its interpretation is identified with its value.

## 4. First-Order and Infinitary Languages

Types, and later Henkin types, are formed from the values that structures assign to formulas. We therefore build, from the symbols of a vocabulary, the terms and the formulas of a first-order language and of its infinitary extension, and interpret them in structures.

For the rest of the paper, we fix a vocabulary $V = (\mathbb{M}, F, P)$.

For each sort descriptor $M \in \mathbb{M}$, we fix a sequence $(x_i^M)_{i<\omega}$ of distinct symbols, not in $V$, chosen so that the sequences for distinct sort descriptors have no symbol in common. Each symbol $x_i^M$ is called a **variable of sort** $M$.

**Terms of sort** $M$ are defined recursively as follows:

- ♠ every variable of sort $M$ is a term of sort $M$;
- ♠ for every function symbol $f\colon \prod_{i<n} M_i \to M$ in $F$ and every tuple $\bar\tau = (\tau_i)_{i<n}$ of terms such that $\tau_i$ is of sort $M_i$ for each $i < n$, the expression $f(\bar\tau)$ is a term of sort $M$.

In particular, every constant symbol of sort $M$ is a term of sort $M$. For a natural number $n$ and a tuple $(M_i)_{i<n}$ of sort descriptors in $\mathbb{M}$, a tuple $(\tau_i)_{i<n}$ of terms such that $\tau_i$ is of sort $M_i$ for each $i < n$ is called a **term of sort** $\prod_{i<n} M_i$; for $n = 0$, the only such term is the empty tuple. A term is **closed** if no variable occurs in it.

For a predicate symbol $p\colon \prod_{i<n} M_i \to [-r_p, r_p]$ in $P$ and a term $\bar\tau$ of sort $\prod_{i<n} M_i$, the expression $p(\bar\tau)$ is called an **atomic $V$-formula** with **a priori bound** $r_{p(\bar\tau)} \coloneqq r_p$. The other $V$-formulas are formed from the atomic ones by applying symbols called **logical connectives**:

- ♠ for each rational number $s$, the symbol $s\cdot$ is a logical connective;
- ♠ the symbol $\vee$ is a logical connective;
- ♠ for each variable $x$, the symbol $\sup_x$ is a logical connective.

The $V$-formulas and their a priori bounds are defined recursively, starting from the clause that every atomic $V$-formula is a $V$-formula. If $\varphi$ and $\psi$ are $V$-formulas with a priori bounds $r_\varphi$ and $r_\psi$, respectively, then the expressions $s\cdot\varphi$, $(\varphi \vee \psi)$ and $\sup_x \varphi$ are $V$-formulas for every rational number $s$ and every variable $x$. The a priori bounds of these three formulas are defined by

$$r_{s\cdot\varphi} \coloneqq |s| r_\varphi, \qquad r_{\varphi\vee\psi} \coloneqq \max\{r_\varphi, r_\psi\}, \qquad r_{\sup_x \varphi} \coloneqq r_\varphi,$$

respectively. Since the connectives $s\cdot$ and $\sup_x$ are written before the formula to which they apply, the parentheses in $(\varphi \vee \psi)$ ensure that every $V$-formula is formed in only one way. Hence the a priori bound of a $V$-formula, like the other notions defined below by recursion on formulas, is well defined; we omit the parentheses when no ambiguity arises. The set of all $V$-formulas is called the **first-order language for** $V$ and is denoted by $L^V$.

The set $\mathrm{free}(\xi)$ of **variables free in a $V$-formula** $\xi$ is defined recursively as follows:

- ♠ if $\xi$ is atomic, of the form $p(\bar\tau)$, then $\mathrm{free}(\xi)$ is the set of variables that occur in the terms of the tuple $\bar\tau$;
- ♠ if $\xi$ is of the form $s\cdot\varphi$, then $\mathrm{free}(\xi) \coloneqq \mathrm{free}(\varphi)$;

- ♠ if $\xi$ is of the form $\varphi \vee \psi$, then $\mathrm{free}(\xi) \coloneqq \mathrm{free}(\varphi) \cup \mathrm{free}(\psi)$;
- ♠ if $\xi$ is of the form $\sup_x \varphi$, then $\mathrm{free}(\xi) \coloneqq \mathrm{free}(\varphi) \setminus \{x\}$.

A $V$**-sentence** is a $V$-formula with no free variables, that is, a $V$-formula $\varphi$ such that $\mathrm{free}(\varphi) = \emptyset$.

For a term $\tau$ and a tuple $\bar{x}$ of distinct variables, we write $\tau(\bar{x})$ to indicate that every variable occurring in $\tau$ is among $\bar{x}$. Similarly, for a formula $\varphi$, we write $\varphi(\bar{x})$ to indicate that every variable free in $\varphi$ is among $\bar{x}$. If $\bar{x} = (x_i)_{i<n}$, where $x_i$ is a variable of sort $M_i$ for each $i < n$, we write

$$_{\bar{x}}M \coloneqq \prod_{i<n} M_i$$

and say that $\bar{x}$ is a variable of sort $_{\bar{x}}M$. For a $V$-structure $\mathfrak{M}$, we write

$$_{\bar{x}}M^{\mathfrak{M}} \coloneqq \prod_{i<n} M_i^{\mathfrak{M}}.$$

The infinitary language $L^V_{\omega_1\omega}$ is obtained by adding to the formation rules of $L^V$ a new logical connective $\bigvee$, called the **infinitary connective of countable join**. More precisely, the language $L^V_{\omega_1\omega}$, together with the a priori bounds and the free variables of its members, is defined by recursion on the countable ordinals. Let $\mathcal{L}_0$ be the set of atomic $V$-formulas, and let $\mathcal{L}_\lambda \coloneqq \bigcup_{\alpha<\lambda} \mathcal{L}_\alpha$ for every countable limit ordinal $\lambda$. For $\alpha < \omega_1$, the set $\mathcal{L}_{\alpha+1}$ is obtained by adding to $\mathcal{L}_\alpha$ two kinds of expressions formed from its members. The first are the expressions $s\cdot\varphi$, $\varphi \vee \psi$ and $\sup_x \varphi$ with $\varphi, \psi \in \mathcal{L}_\alpha$, for all rational numbers $s$ and all variables $x$; their a priori bounds and free variables are given by the clauses stated for $L^V$. The second concern the sequences $\Phi = (\varphi_n)_{n<\omega}$ of members of $\mathcal{L}_\alpha$ for which $\sup_{n<\omega} r_{\varphi_n}$ is finite and, for some finite tuple $\bar{x}$ of variables, every $\varphi_n$ is of the form $\varphi_n(\bar{x})$. For every such sequence, the expression $\bigvee \Phi$ is added, with a priori bound $r_{\bigvee\Phi} \coloneqq \sup_{n<\omega} r_{\varphi_n}$. The free variables of $\bigvee \Phi$ are those of the formulas $\varphi_n$, that is, $\mathrm{free}(\bigvee \Phi) \coloneqq \bigcup_{n<\omega} \mathrm{free}(\varphi_n)$, a subset of the finite set of variables of $\bar{x}$. Finally, we set $L^V_{\omega_1\omega} \coloneqq \bigcup_{\alpha<\omega_1} \mathcal{L}_\alpha$. Since $\omega_1$ is regular, every sequence $(\varphi_n)_{n<\omega}$ of members of $L^V_{\omega_1\omega}$ consists of members of some $\mathcal{L}_\alpha$, and the set $L^V_{\omega_1\omega}$ is therefore closed under all these formation rules.

We also write $\bigvee_n \varphi_n$ for $\bigvee \Phi$, and we denote the set of sentences of $L^V_{\omega_1\omega}$ by $S^V_{\omega_1\omega}$. A formula of $L^V_{\omega_1\omega}$ may have infinite length, but only finitely many free variables. The two languages differ in size: if $V$ is countable, the first-order language $L^V$ is countable, whereas the infinitary language $L^V_{\omega_1\omega}$ has the cardinality of the continuum. For this reason, the Baire category argument of this paper takes place in countable fragments (see Section 6).

Let $\mathfrak{M}$ be a $V$-structure, and let $\tau(\bar{x})$ be a term of sort $M$, where $\bar{x} = (x_j)_{j<m}$. The **interpretation of** $\tau(\bar{x})$ **in** $\mathfrak{M}$ is the function $\tau^{\mathfrak{M}} \colon {}_{\bar{x}}M^{\mathfrak{M}} \to M^{\mathfrak{M}}$ defined recursively, for $\bar{a} = (a_j)_{j<m}$ in $_{\bar{x}}M^{\mathfrak{M}}$, as follows:

- ♠ the interpretation of the variable $x_j$ is the projection $x_j^{\mathfrak{M}}(\bar{a}) \coloneqq a_j$;

- ♠ if $f\colon \prod_{i<n} M_i \to M$ is a function symbol and $\bar\tau = (\tau_i)_{i<n}$ is a term of sort $\prod_{i<n} M_i$, then the interpretation of the term $f(\bar\tau)$ is given by $(f(\bar\tau))^{\mathfrak{M}}(\bar a) \coloneqq f^{\mathfrak{M}}((\tau_i^{\mathfrak{M}}(\bar a))_{i<n})$.

The interpretation of a closed term $\tau$ is a function on the one-point set ${}_{\bar x}M^{\mathfrak{M}}$ with $\bar x$ empty, and is identified with its value, an element of $M^{\mathfrak{M}}$. The **interpretation of a formula** $\xi(\bar x)$ **of** $L^V_{\omega_1\omega}$ **in** $\mathfrak{M}$ is the function $\xi^{\mathfrak{M}}$ on ${}_{\bar x}M^{\mathfrak{M}}$ defined recursively, for $\bar a$ in ${}_{\bar x}M^{\mathfrak{M}}$, as follows:

- ♠ if $\xi$ is atomic, of the form $p(\bar\tau)$ with $\bar\tau = (\tau_i)_{i<n}$, then
$$\xi^{\mathfrak{M}}(\bar a) \coloneqq p^{\mathfrak{M}}((\tau_i^{\mathfrak{M}}(\bar a))_{i<n});$$
- ♠ if $\xi$ is of the form $s\cdot\varphi$, where $s$ is a rational number and $\varphi \in L^V_{\omega_1\omega}$, then
$$\xi^{\mathfrak{M}}(\bar a) \coloneqq s\cdot\varphi^{\mathfrak{M}}(\bar a);$$
- ♠ if $\xi$ is of the form $\varphi\vee\psi$, where $\varphi,\psi \in L^V_{\omega_1\omega}$, then
$$\xi^{\mathfrak{M}}(\bar a) \coloneqq \max\{\varphi^{\mathfrak{M}}(\bar a), \psi^{\mathfrak{M}}(\bar a)\};$$
- ♠ if $\xi$ is of the form $\sup_y \varphi$, where $\varphi \in L^V_{\omega_1\omega}$, then
$$\xi^{\mathfrak{M}}(\bar a) \coloneqq \sup_{b\in {}_yM^{\mathfrak{M}}} \varphi^{\mathfrak{M}}(\bar a[b/y]),$$
where, if $y$ is not among $\bar x$, the formula $\varphi$ is regarded as a formula $\varphi(\bar x y)$, and $\bar a[b/y]$ is the tuple $\bar a b$; if $y$ is the variable $x_j$ of $\bar x$, the formula $\varphi$ is regarded as a formula $\varphi(\bar x)$, and $\bar a[b/y]$ is the tuple obtained from $\bar a$ by replacing its entry $a_j$ by $b$;
- ♠ if $\xi$ is of the form $\bigvee\Phi$, where $\Phi = (\varphi_n(\bar x))_{n<\omega}$ is a sequence of formulas of $L^V_{\omega_1\omega}$, then
$$\xi^{\mathfrak{M}}(\bar a) \coloneqq \sup_{n<\omega} \varphi_n^{\mathfrak{M}}(\bar a).$$

An induction on formulas shows that the suprema in this definition are finite and that $\xi^{\mathfrak{M}}$ takes its values in $[-r_\xi, r_\xi]$. When $\xi$ is a sentence, the tuple $\bar x$ may be taken empty, and $\xi^{\mathfrak{M}}$ is identified with its value, a real number in $[-r_\xi, r_\xi]$.

## 5. Expansion by Constants

In the method introduced by Henkin in 1949, the elements of a structure are named by new constant symbols; the first step is to add these symbols to the fixed vocabulary. Let $C = \{c_n^M : n<\omega,\ M\in\mathbb{M}\}$ be a set of distinct constant symbols, not in $V$, such that $c_n^M$ is of sort $M$ for every $n<\omega$ and every $M\in\mathbb{M}$. The **expansion of** $V$ **by** $C$ is the vocabulary $V_C \coloneqq (\mathbb{M}, F\cup C, P)$, which is countable if $V$ is; the infinitary language $L^{V_C}_{\omega_1\omega}$ is denoted by $L^C_{\omega_1\omega}$, and its set of sentences by $S^C_{\omega_1\omega}$.

Given a $V$-structure $\mathfrak{M} = \langle (M^{\mathfrak{M}})_{M\in\mathbb{M}}, (f^{\mathfrak{M}})_{f\in F}, (p^{\mathfrak{M}})_{p\in P}\rangle$, an **expansion of** $\mathfrak{M}$ **to** $V_C$ is a $V_C$-structure
$$\mathfrak{N} = \langle (M^{\mathfrak{N}})_{M\in\mathbb{M}}, (f^{\mathfrak{N}})_{f\in F\cup C}, (p^{\mathfrak{N}})_{p\in P}\rangle$$

that agrees with $\mathfrak{M}$ on the symbols of $V$, in the sense that $M^{\mathfrak{N}} = M^{\mathfrak{M}}$ for every $M \in \mathbb{M}$, that $f^{\mathfrak{N}} = f^{\mathfrak{M}}$ for every $f \in F$, and that $p^{\mathfrak{N}} = p^{\mathfrak{M}}$ for every $p \in P$. Each expansion $\mathfrak{N}$ of $\mathfrak{M}$ to $V_C$ is determined by the elements $(c_n^M)^{\mathfrak{N}}$ of $M^{\mathfrak{M}}$ that interpret the new constant symbols. Conversely, every $V_C$-structure $\mathfrak{N}$ is an expansion of exactly one $V$-structure, its **$V$-reduct**, obtained by forgetting the interpretations of the symbols in $C$.

In the structures built by the Henkin construction, every element has a name. A $V_C$-structure $\mathfrak{N}$ is **canonical** if, for every sort descriptor $M \in \mathbb{M}$, every element of $M^{\mathfrak{N}}$ is the interpretation in $\mathfrak{N}$ of a closed $V_C$-term of sort $M$.

## 6. Fragments

The Baire category argument requires the Henkin types of Section 9 to form a $G_\delta$ subset of a compact space. In Section 10 we write these types as an intersection of open sets attached to sentences, an intersection that is countable only when the sentences are countable in number. The infinitary language, however, has the cardinality of the continuum even when the vocabulary is countable (see Section 4). The argument is therefore carried out in fragments, sets of formulas closed under the first-order connectives, under subformulas and under substitution of terms. A fragment is countable when it is generated by countably many formulas over a countable vocabulary.

Let $\xi$ be a formula of $L^V_{\omega_1\omega}$. The set $\mathrm{sub}(\xi)$ of **subformulas of** $\xi$ is defined recursively as follows:

- ♠ if $\xi$ is atomic, we set $\mathrm{sub}(\xi) \coloneqq \{\xi\}$;
- ♠ if $\xi$ is of the form $s\cdot\varphi$, we set $\mathrm{sub}(\xi) \coloneqq \{\xi\} \cup \mathrm{sub}(\varphi)$;
- ♠ if $\xi$ is of the form $\varphi \vee \psi$, we set $\mathrm{sub}(\xi) \coloneqq \{\xi\} \cup \mathrm{sub}(\varphi) \cup \mathrm{sub}(\psi)$;
- ♠ if $\xi$ is of the form $\sup_x \varphi$, we set $\mathrm{sub}(\xi) \coloneqq \{\xi\} \cup \mathrm{sub}(\varphi)$;
- ♠ if $\xi$ is of the form $\bigvee_n \varphi_n$, we set $\mathrm{sub}(\xi) \coloneqq \{\xi\} \cup \bigcup_{n<\omega} \mathrm{sub}(\varphi_n)$.

A **fragment** of $L^V_{\omega_1\omega}$ is a set $\mathcal{F} \subseteq L^V_{\omega_1\omega}$ with the following properties:

- (F1) it contains $L^V$;
- (F2) it is closed under subformulas, that is, $\mathrm{sub}(\varphi) \subseteq \mathcal{F}$ for every $\varphi \in \mathcal{F}$;
- (F3) it is closed under the first-order connectives, that is, if $\varphi, \psi \in \mathcal{F}$, then $s\cdot\varphi \in \mathcal{F}$ for every rational number $s$, $\varphi \vee \psi \in \mathcal{F}$, and $\sup_x \varphi \in \mathcal{F}$ for every variable $x$;
- (F4) it is closed under substitution of terms, that is, if $\varphi(\bar{x}y) \in \mathcal{F}$ and $\tau$ is a term of sort ${}_yM$ that is free for $y$ in $\varphi$ (no free occurrence of $y$ in $\varphi$ lies within a subformula $\sup_z \psi$ of $\varphi$ such that $z$ occurs in $\tau$), then the formula $\varphi(\bar{x}\tau)$, obtained from $\varphi$ by substituting $\tau$ for the free occurrences of $y$, belongs to $\mathcal{F}$.

The intersection of a nonempty family of fragments of $L^V_{\omega_1\omega}$ is a fragment, and the language $L^V_{\omega_1\omega}$ is itself a fragment. Hence, for every set $\Phi \subseteq L^V_{\omega_1\omega}$, the intersection of all fragments containing $\Phi$ is the smallest fragment containing $\Phi$, which is called the **fragment generated by** $\Phi$. If $V$ and $\Phi$ are countable, then the fragment generated by $\Phi$ is countable. Indeed, let $\mathcal{F}_0 \coloneqq L^V \cup \bigcup_{\varphi\in\Phi} \mathrm{sub}(\varphi)$; for

each $k < \omega$, let $\mathcal{F}_{k+1}$ be obtained by adding to $\mathcal{F}_k$ the formulas that (F2)–(F4) require for members of $\mathcal{F}_k$. Since every formula has countably many subformulas, and since there are countably many rational numbers, variables and terms, an induction on $k$ shows that every set $\mathcal{F}_k$ is countable. The union of the increasing sequence $(\mathcal{F}_k)_{k<\omega}$ is a fragment that contains $\Phi$ and is contained in every fragment containing $\Phi$. Consequently, the fragment generated by $\Phi$ is this union, which is countable.

Let $\mathcal{F}$ be a fragment of $L^V_{\omega_1\omega}$. Since $L^V_{\omega_1\omega}$ is contained in the infinitary language $L^C_{\omega_1\omega}$ of the expanded vocabulary $V_C$ of Section 5, we may consider the fragment of $L^C_{\omega_1\omega}$ generated by $\mathcal{F}$, which we denote by $\mathcal{F}_C$. If $V$ and $\mathcal{F}$ are countable, then so is $\mathcal{F}_C$, by the preceding paragraph applied to the countable vocabulary $V_C$. We denote the set of sentences of $\mathcal{F}$ by $S^{\mathcal{F}}$ and the set of sentences of $\mathcal{F}_C$ by $S^{\mathcal{F}}_C$.

## 7. Conditions and Theories

In real-valued logic, the role of the statements of classical logic is played by conditions, formal strict inequalities between a sentence and a rational number. Sets of conditions, called theories, specify the models to be constructed, whereas the types to be omitted are expressed by closed conditions. Strict conditions are taken as the basic ones because each of them defines an open set in the topology of Section 10, as the Baire category argument requires.

A **lower-bound condition** is a formal expression $\varphi > r$, and an **upper-bound condition** is a formal expression $\varphi < r$. Each of these expressions is formed from a sentence $\varphi$ of the infinitary language $L^C_{\omega_1\omega}$ of the expanded vocabulary $V_C$ (see Section 5) and a rational number $r$. A **condition** is a lower-bound condition or an upper-bound condition, and a **theory** is a set of conditions. For a set $\Sigma$ of sentences, a **$\Sigma$-theory** is a theory whose conditions are on sentences in $\Sigma$.

For a rational number $r$, we write

- ♠ $\varphi \geq r$ for the countable theory $\{\varphi > r - \frac{1}{n} : n \in \mathbb{Z}_+\}$, and
- ♠ $\varphi \leq r$ for the countable theory $\{\varphi < r + \frac{1}{n} : n \in \mathbb{Z}_+\}$.

The expressions $\varphi \geq r$ and $\varphi \leq r$ are called **closed conditions**.

Let $\mathfrak{N}$ be a $V_C$-structure. We say that $\mathfrak{N}$ **satisfies** the condition $\varphi > r$, and write $\mathfrak{N} \models \varphi > r$, if $\varphi^{\mathfrak{N}} > r$; it satisfies the condition $\varphi < r$ if $\varphi^{\mathfrak{N}} < r$. A $V_C$-structure that satisfies every condition of a theory is called a **model** of the theory; the same term is used for $V$-structures and theories whose conditions are on $V$-sentences. Thus a structure satisfies the closed condition $\varphi \geq r$ exactly when the value of $\varphi$ in it is at least $r$.

Under the reading of $1$ as true and $-1$ as false, the connectives generalize those of classical first-order logic, as the following equivalences show. Each of these equivalences holds for every $V_C$-structure $\mathfrak{N}$ and every rational number $r$. Multiplication by $-1$ plays the role of negation: for a sentence $\varphi$, if $-\varphi$ denotes the

sentence obtained by applying the connective $s\cdot$ with $s = -1$, then

$$\begin{aligned} \mathfrak{N} \models -\varphi > r \quad &\text{iff} \quad (-\varphi)^{\mathfrak{N}} > r \\ &\text{iff} \quad -(\varphi^{\mathfrak{N}}) > r \\ &\text{iff} \quad \varphi^{\mathfrak{N}} < -r. \end{aligned}$$

In particular, for $r = 0$, a $V_C$-structure satisfies $-\varphi > 0$ exactly when it satisfies $\varphi < 0$. The binary connective $\vee$ is the counterpart of disjunction: for sentences $\varphi$ and $\psi$, we have

$$\begin{aligned} \mathfrak{N} \models \varphi \vee \psi > r \quad &\text{iff} \quad \max\{\varphi^{\mathfrak{N}}, \psi^{\mathfrak{N}}\} > r \\ &\text{iff} \quad \mathfrak{N} \models \varphi > r \text{ or } \mathfrak{N} \models \psi > r. \end{aligned}$$

The supremum $\sup_x$ takes the place of the existential quantifier: for a formula $\varphi(x)$ of $L^C_{\omega_1\omega}$, the condition $\sup_x \varphi > r$ is characterized as follows:

$$\begin{aligned} \mathfrak{N} \models \sup_x \varphi > r \quad &\text{iff} \quad (\sup_x \varphi)^{\mathfrak{N}} > r \\ &\text{iff} \quad \sup\left\{\varphi^{\mathfrak{N}}(a) \colon a \in {}_xM^{\mathfrak{N}}\right\} > r \\ &\text{iff} \quad \varphi^{\mathfrak{N}}(a) > r \text{ for some } a \in {}_xM^{\mathfrak{N}}. \end{aligned}$$

Finally, the join $\bigvee$ generalizes countable disjunction: for a sequence $\Phi$ of sentences of $L^C_{\omega_1\omega}$ whose join is a sentence of $L^C_{\omega_1\omega}$, the condition $\bigvee \Phi > r$ is characterized in the same way:

$$\begin{aligned} \mathfrak{N} \models \bigvee \Phi > r \quad &\text{iff} \quad \left(\bigvee \Phi\right)^{\mathfrak{N}} > r \\ &\text{iff} \quad \sup\left\{\varphi^{\mathfrak{N}} \colon \varphi \in \Phi\right\} > r \\ &\text{iff} \quad \varphi^{\mathfrak{N}} > r \text{ for some } \varphi \in \Phi. \end{aligned}$$

## 8. Types of Structures and Formal Types

The type of a structure records the values that the structure assigns to a set of sentences; formal types are families of such values that need not come from any structure. Among the latter are the Henkin types, which, as we show in Section 9, do come from structures. The Baire category argument takes place in the space formed by the Henkin types, a subspace of the space of formal types with the product topology.

Let $\Phi$ be a set of sentences of the infinitary language $L^C_{\omega_1\omega}$ of the expanded vocabulary $V_C$. We write $\prod \Phi$ for the product $\prod_{\varphi\in\Phi}[-r_\varphi, r_\varphi]$ endowed with the product topology, where $r_\varphi$ is the a priori bound of $\varphi$. In this topology, the product $\prod \Phi$ is a compact Hausdorff space (see Section 2); its elements are called **formal $\Phi$-types**. For a formal $\Phi$-type $\mathfrak{m}$ and a sentence $\varphi \in \Phi$, we write $\varphi^{\mathfrak{m}}$ for the $\varphi$-coordinate of $\mathfrak{m}$ and say that $\mathfrak{m}$ satisfies a condition $\varphi > r$ or $\varphi < r$ if $\varphi^{\mathfrak{m}} > r$ or $\varphi^{\mathfrak{m}} < r$, respectively. If $\mathfrak{N}$ is a $V_C$-structure, the **$\Phi$-type of** $\mathfrak{N}$ is the formal $\Phi$-type

$$\Phi\text{-tp}(\mathfrak{N}) \coloneqq (\varphi^{\mathfrak{N}})_{\varphi\in\Phi}.$$

When $\Phi$ is contained in the set $S^V_{\omega_1\omega}$ of sentences of $L^V_{\omega_1\omega}$, the same definition applies to a $V$-structure $\mathfrak{M}$. The $S^V_{\omega_1\omega}$-type $S^V_{\omega_1\omega}\text{-tp}(\mathfrak{M})$ is called the **full type** of $\mathfrak{M}$; similarly, the full type of a $V_C$-structure is its type for the set $S^C_{\omega_1\omega}$ of sentences of $L^C_{\omega_1\omega}$.

## 9. Henkin Types

In his proof of the completeness theorem, Henkin [Hen49] built a model of a consistent theory out of new individual constants, chosen to provide every existential sentence of the theory with a witness among them. The Henkin conditions below are the real-valued form of this requirement. They demand of a formal type that the value of a sentence formed with a connective from other sentences be computed from their values. The value of a supremum over a variable must likewise be the supremum of the values obtained by substituting closed terms for that variable. In this section we show that the formal types satisfying the Henkin conditions are exactly the types of canonical structures, structures in which every element is the value of a closed term. We also deduce, with the help of a downward Löwenheim–Skolem theorem for countable fragments, which formal types are realized by structures.

Let $\mathcal{G}$ be a fragment of the infinitary language $L^C_{\omega_1\omega}$ of the expanded vocabulary $V_C$, and let $S^{\mathcal{G}}$ be the set of its sentences. Let $x$ be a variable, let $\varphi$ be a formula whose free variables are among $x$, and let $\tau$ be a closed $V_C$-term of sort ${}_xM$. We write $\varphi(\tau)$ for the sentence obtained by substituting $\tau$ for the free occurrences of $x$ in $\varphi$, as in condition (F4) of the definition of a fragment (see Section 6). A formal type $\mathfrak{m} \in \prod S^{\mathcal{G}}$ is an $S^{\mathcal{G}}$**-Henkin type** if it satisfies the following **Henkin conditions**:

- (H1) $1^{\mathfrak{m}} = 1$;
- (H2) $(s\cdot\varphi)^{\mathfrak{m}} = s\,\varphi^{\mathfrak{m}}$ for every $\varphi \in S^{\mathcal{G}}$ and every $s \in \mathbb{Q}$;
- (H3) $(\bigvee \Phi)^{\mathfrak{m}} = \sup\{\varphi^{\mathfrak{m}} \colon \varphi \in \Phi\}$ for every sentence $\bigvee \Phi \in S^{\mathcal{G}}$;
- (H4) $(\sup_x \varphi)^{\mathfrak{m}} = \sup\{\varphi(\tau)^{\mathfrak{m}} \colon \tau \text{ is a closed } V_C\text{-term of sort } {}_xM\}$ for every variable $x$ and every formula $\varphi \in \mathcal{G}$ whose free variables are among $x$;
- (H5) $(\varphi \vee \psi)^{\mathfrak{m}} = \max\{\varphi^{\mathfrak{m}}, \psi^{\mathfrak{m}}\}$ for all $\varphi, \psi \in S^{\mathcal{G}}$.

Every sentence that occurs in these conditions belongs to $S^{\mathcal{G}}$: the logical unity constant $1$ by (F1), and the sentences $s\cdot\varphi$, $\varphi \vee \psi$ and $\sup_x \varphi$ by (F3). The members of $\Phi$ belong to $S^{\mathcal{G}}$ by (F2), since the free variables of $\bigvee \Phi$ are those of its members; the sentences $\varphi(\tau)$ belong to $S^{\mathcal{G}}$ by (F4). The language $L^C_{\omega_1\omega}$ is a fragment of itself, whose set of sentences is $S^C_{\omega_1\omega}$; its $S^C_{\omega_1\omega}$-Henkin types lie in $\prod S^C_{\omega_1\omega}$ and are simply called **Henkin types**.

To pass from formal types back to structures, we follow Henkin and take as elements the closed terms themselves; the structure so obtained depends only on the values of the atomic sentences. Let $\mathcal{A}$ be the set of atomic $V_C$-sentences; by (F1), it is contained in $S^{\mathcal{G}}$ for every fragment $\mathcal{G}$ of $L^C_{\omega_1\omega}$. An **atomic type** is a formal type $\mathfrak{a} \in \prod \mathcal{A}$ such that $1^{\mathfrak{a}} = 1$. Every atomic type $\mathfrak{a}$ determines a $V_C$-structure $\mathfrak{N}_{\mathfrak{a}}$, the **term structure of** $\mathfrak{a}$, as follows. For each $M \in \mathbb{M}$, the set $M^{\mathfrak{N}_{\mathfrak{a}}}$ consists of

the closed $V_C$-terms of sort $M$; it is nonempty, since it contains the constants $c_n^M$. Each function symbol $f$ of $V_C$ is interpreted syntactically, by $f^{\mathfrak{N}_\mathfrak{a}}(\bar\tau) := f(\bar\tau)$, and each predicate symbol $p$ by $p^{\mathfrak{N}_\mathfrak{a}}(\bar\tau) := p(\bar\tau)^\mathfrak{a}$. The interpretation $p^{\mathfrak{N}_\mathfrak{a}}$ is a function into $[-r_p, r_p]$, because the a priori bound of the atomic sentence $p(\bar\tau)$ is $r_p$. Since, moreover, the logical unity constant satisfies $1^{\mathfrak{N}_\mathfrak{a}} = 1^\mathfrak{a} = 1$, the term structure $\mathfrak{N}_\mathfrak{a}$ is a $V_C$-structure. No quotient of the set of terms is needed, since the vocabulary has no equality symbol. An induction on terms shows that $\tau^{\mathfrak{N}_\mathfrak{a}} = \tau$ for every closed $V_C$-term $\tau$; hence the term structure $\mathfrak{N}_\mathfrak{a}$ is canonical and satisfies $\mathcal{A}\text{-tp}(\mathfrak{N}_\mathfrak{a}) = \mathfrak{a}$.

**Lemma 2.** Let $\mathcal{G}$ be a fragment of $L^C_{\omega_1\omega}$.

(a) If $\mathfrak{N}$ is a canonical $V_C$-structure, then $S^\mathcal{G}\text{-tp}(\mathfrak{N})$ is an $S^\mathcal{G}$-Henkin type.
(b) Two $S^\mathcal{G}$-Henkin types that agree on $\mathcal{A}$ are equal.
(c) A formal type $\mathfrak{m} \in \prod S^\mathcal{G}$ is the $S^\mathcal{G}$-type of a canonical $V_C$-structure if and only if it is an $S^\mathcal{G}$-Henkin type; in that case, it is the $S^\mathcal{G}$-type of the term structure of its restriction to $\mathcal{A}$.

*Proof.* The idea is that the Henkin conditions repeat the recursive definition of the values of sentences, except that a supremum over a sort becomes a supremum over closed terms. The Henkin conditions therefore determine the value of every sentence from the values of the atomic ones, which is part (b). Part (a) asserts that the same conditions hold in a canonical structure, in which every element is the value of a closed term and the two suprema coincide. Part (b) is proved by induction on the rank of formulas: atomic formulas have rank 0, and the rank of $s\cdot\varphi$ and of $\sup_x \varphi$ is the rank of $\varphi$ plus one. The rank of $\varphi \vee \psi$ is the larger of the ranks of $\varphi$ and $\psi$ plus one. Finally, the rank of $\bigvee \Phi$ is the supremum of the ranks of the members of $\Phi$ plus one. Substituting closed terms for variables changes only the atomic subformulas, and therefore preserves the rank.

(a) Conditions (H1), (H2), (H3) and (H5) hold for the values $\sigma^\mathfrak{N}$ of the sentences $\sigma \in S^\mathcal{G}$, by the definition of a $V_C$-structure and the recursive definition of the interpretation of formulas. For (H4), let $\varphi \in \mathcal{G}$ be a formula whose free variables are among $x$. The substitution lemma asserts that $\psi(\bar z\tau)^\mathfrak{N}(\bar b) = \psi^\mathfrak{N}(\bar b\, \tau^\mathfrak{N})$ for every formula $\psi(\bar z x)$ of $L^C_{\omega_1\omega}$, where $x$ is not among $\bar z$, every closed $V_C$-term $\tau$ of sort ${}_xM$ and every tuple $\bar b \in {}_{\bar z}M^\mathfrak{N}$. It is proved by a routine induction on $\psi$, which rests on the corresponding identity for terms and needs the parameters $\bar z$ in the step for suprema; here we use the case in which $\bar z$ is empty and $\psi$ is $\varphi$. Since $\mathfrak{N}$ is canonical, every element of ${}_xM^\mathfrak{N}$ is of the form $\tau^\mathfrak{N}$ for a closed $V_C$-term $\tau$ of sort ${}_xM$; this gives

$$(\sup_x \varphi)^\mathfrak{N} = \sup_{b\in {}_xM^\mathfrak{N}} \varphi^\mathfrak{N}(b) = \sup_\tau \varphi^\mathfrak{N}(\tau^\mathfrak{N}) = \sup_\tau \varphi(\tau)^\mathfrak{N}.$$

(b) Let $\mathfrak{m}$ and $\mathfrak{m}'$ be $S^\mathcal{G}$-Henkin types that agree on $\mathcal{A}$. We show by induction on the rank of $\sigma$ that $\sigma^\mathfrak{m} = \sigma^{\mathfrak{m}'}$ for every $\sigma \in S^\mathcal{G}$, the atomic case being the hypothesis. Suppose that $\sigma$ is of the form $s\cdot\varphi$, $\varphi\vee\psi$ or $\bigvee\Phi$. In that case, the formulas $\varphi$, $\psi$ and the members of $\Phi$ are sentences in $S^\mathcal{G}$ of smaller rank, on which the types $\mathfrak{m}$ and $\mathfrak{m}'$ agree by the induction hypothesis. Conditions (H2), (H5) and

(H3), respectively, compute $\sigma^{\mathfrak{m}}$ and $\sigma^{\mathfrak{m}'}$ alike from the values of these sentences. Suppose, finally, that $\sigma$ is of the form $\sup_x \varphi$. For every closed $V_C$-term $\tau$ of sort ${}_xM$, the sentence $\varphi(\tau)$ then belongs to $S^{\mathcal{G}}$ and has smaller rank. The types $\mathfrak{m}$ and $\mathfrak{m}'$ agree on each of these sentences for the same reason, and condition (H4) therefore gives $\sigma^{\mathfrak{m}} = \sigma^{\mathfrak{m}'}$.

(c) By (a), the $S^{\mathcal{G}}$-type of a canonical $V_C$-structure is an $S^{\mathcal{G}}$-Henkin type. Conversely, let $\mathfrak{m}$ be an $S^{\mathcal{G}}$-Henkin type, and let $\mathfrak{a}$ be its restriction to $\mathcal{A}$, an atomic type by (H1). The term structure $\mathfrak{N}_{\mathfrak{a}}$ is canonical, with $\mathcal{A}\text{-tp}(\mathfrak{N}_{\mathfrak{a}}) = \mathfrak{a}$. By (a), the $S^{\mathcal{G}}$-type of $\mathfrak{N}_{\mathfrak{a}}$ is an $S^{\mathcal{G}}$-Henkin type that agrees with $\mathfrak{m}$ on $\mathcal{A}$; by (b), the two types are equal. □

Let $\mathfrak{a}$ be an atomic type. By Lemma 2(a), applied to $L^C_{\omega_1\omega}$, the full type $S^C_{\omega_1\omega}\text{-tp}(\mathfrak{N}_{\mathfrak{a}})$ is a Henkin type that extends $\mathfrak{a}$. Part (b) of that lemma shows that no other Henkin type extends $\mathfrak{a}$. We call this full type the **Henkin type of** $\mathfrak{a}$ and denote it by $\hat{\mathfrak{a}}$. For every fragment $\mathcal{G}$ of $L^C_{\omega_1\omega}$, the same lemma shows that the restriction of $\hat{\mathfrak{a}}$ to $S^{\mathcal{G}}$ is the only $S^{\mathcal{G}}$-Henkin type that extends $\mathfrak{a}$.

Let $\mathfrak{N}$ be a $V_C$-structure, and let $\mathfrak{a} \coloneqq \mathcal{A}\text{-tp}(\mathfrak{N})$. If $\mathfrak{N}$ is canonical, then Lemma 2(a) shows that the full type $S^C_{\omega_1\omega}\text{-tp}(\mathfrak{N})$ is a Henkin type that extends $\mathfrak{a}$, and therefore equals $\hat{\mathfrak{a}}$. In other words, we have $\varphi^{\mathfrak{N}} = \varphi^{\hat{\mathfrak{a}}}$ for every $\varphi \in S^C_{\omega_1\omega}$. If $\mathfrak{N}$ is not canonical, the two values may differ, and in either direction. For example, suppose that $V$ has a single sort descriptor $M$ and no function symbols, and that the only predicate symbol of $V$ besides $1$ is $p\colon M \to [-1,1]$. Let $\mathfrak{N}$ be the $V_C$-structure with $M^{\mathfrak{N}} = \{0,1\}$, $p^{\mathfrak{N}}(0) = -1$ and $p^{\mathfrak{N}}(1) = 1$, in which every constant of $C$ is interpreted as $0$. For a variable $x$ of sort $M$, the sentence $\sup_x p(x)$ takes the value $1$ in $\mathfrak{N}$, whereas $(\sup_x p(x))^{\hat{\mathfrak{a}}} = \sup_n p(c^M_n)^{\mathfrak{a}} = -1$, the constants $c^M_n$ being the only closed $V_C$-terms. Applying the connective $s\cdot$ with $s = -1$ to this sentence reverses the inequality.

Let $\Phi$ be a subset of the set $S^V_{\omega_1\omega}$ of sentences of $L^V_{\omega_1\omega}$, and let $\mathfrak{m} \in \prod \Phi$ be given. We say that the formal type $\mathfrak{m}$ is **realized** if there is a $V$-structure $\mathfrak{M}$ such that $\Phi\text{-tp}(\mathfrak{M}) = \mathfrak{m}$; in that case, we say that $\mathfrak{M}$ **realizes** $\mathfrak{m}$. When $V$ is countable, there are countably many closed $V_C$-terms. In that case, a $V$-structure can therefore have a canonical expansion to $V_C$, to which Lemma 2 applies, only if its sorts are countable. The following form of the downward Löwenheim–Skolem theorem replaces every $V$-structure by one whose sorts are countable and which assigns the same values to the sentences of a given countable fragment.

**Lemma 3.** Assume that the vocabulary $V$ is countable, and let $\mathcal{F}$ be a countable fragment of $L^V_{\omega_1\omega}$. For every $V$-structure $\mathfrak{M}$ there is a $V$-structure $\mathfrak{M}_0$ whose sorts are countable such that $\varphi^{\mathfrak{M}_0} = \varphi^{\mathfrak{M}}$ for every $\varphi$ in the set $S^{\mathcal{F}}$ of sentences of $\mathcal{F}$.

*Proof.* The structure $\mathfrak{M}_0$ is obtained by closing countably many elements of $\mathfrak{M}$ under the functions of $\mathfrak{M}$ and under the choice of approximate witnesses for the suprema of formulas of $\mathcal{F}$. Let $\psi \in \mathcal{F}$, let $y$ be a variable, and let $\bar{x}$ be a finite tuple of distinct variables among which are the free variables of $\sup_y \psi$. For every tuple $\bar{a} \in {}_{\bar{x}}M^{\mathfrak{M}}$ and every integer $k \geq 1$, choose an element $b_{\psi,y,\bar{x},\bar{a},k} \in {}_yM^{\mathfrak{M}}$ such that $\psi^{\mathfrak{M}}(\bar{a}[b_{\psi,y,\bar{x},\bar{a},k}/y]) > (\sup_y \psi)^{\mathfrak{M}}(\bar{a}) - 1/k$, with the notation of the clause for

suprema in Section 4. For each $M \in \mathbb{M}$, let $A_0^M$ consist of one element of $M^{\mathfrak{M}}$. For each $j < \omega$ and each $M \in \mathbb{M}$, let $A_{j+1}^M$ consist of the elements of $A_j^M$, the values in $M^{\mathfrak{M}}$ of the functions $f^{\mathfrak{M}}$ at tuples from $A_j$, and the witnesses $b_{\psi,y,\bar{x},\bar{a},k}$ in $M^{\mathfrak{M}}$ with $\bar{a}$ from $A_j$. Here a tuple is *from* $A_j$ if each of its entries lies in the set $A_j^N$ of its sort $N$. Since $V$ and $\mathcal{F}$ are countable, and since there are countably many variables and finite tuples of variables, an induction on $j$ shows that every set $A_j^M$ is countable; hence so is $A^M \coloneqq \bigcup_j A_j^M$. Moreover, the sets $A_j^M$ increase with $j$, whence the family $(A^M)_{M\in\mathbb{M}}$ is closed under the functions $f^{\mathfrak{M}}$. Let $\mathfrak{M}_0$ be the $V$-structure whose sorts are the nonempty sets $A^M$ and whose functions and predicates are the restrictions of those of $\mathfrak{M}$. By induction on the rank of $\varphi \in \mathcal{F}$ (see the proof of Lemma 2), we show that $\varphi^{\mathfrak{M}_0}(\bar{a}) = \varphi^{\mathfrak{M}}(\bar{a})$ for every tuple $\bar{a}$ from $\mathfrak{M}_0$ of the appropriate sort. For atomic $\varphi$ this holds because terms take the same values in $\mathfrak{M}_0$ and in $\mathfrak{M}$. The steps for $s\cdot$, $\vee$ and $\bigvee$ follow from the definitions, the subformulas of members of $\mathcal{F}$ being members of $\mathcal{F}$ by (F2). Let $\varphi$ be of the form $\sup_y \psi$, and let $\bar{a}$ be a tuple from $\mathfrak{M}_0$ of sort ${}_{\bar{x}}M$, where $\bar{x}$ is a finite tuple of distinct variables among which are the free variables of $\varphi$. The induction hypothesis gives $\varphi^{\mathfrak{M}_0}(\bar{a}) \leq \varphi^{\mathfrak{M}}(\bar{a})$, since the supremum defining $\varphi^{\mathfrak{M}_0}(\bar{a})$ ranges over a subset of ${}_yM^{\mathfrak{M}}$. Conversely, the tuple $\bar{a}$ is from $A_j$ for some $j$. For every $k \geq 1$, the witness $b \coloneqq b_{\psi,y,\bar{x},\bar{a},k}$ is therefore an element of $\mathfrak{M}_0$, and the induction hypothesis applied to $\psi$ gives $\varphi^{\mathfrak{M}_0}(\bar{a}) \geq \psi^{\mathfrak{M}_0}(\bar{a}[b/y]) = \psi^{\mathfrak{M}}(\bar{a}[b/y]) > \varphi^{\mathfrak{M}}(\bar{a}) - 1/k$. The case of sentences is the lemma. □

**Proposition 4** (Henkin Realizability)**.** Assume that the vocabulary $V$ is countable, and let $\Phi \subseteq S^V_{\omega_1\omega}$ and $\mathfrak{m} \in \prod \Phi$ be given.

(a) The formal type $\mathfrak{m}$ is realized by a $V$-structure whose sorts are countable if and only if it extends to a Henkin type in $\prod S^C_{\omega_1\omega}$.
(b) If $\Phi \subseteq S^{\mathcal{F}}$ for a countable fragment $\mathcal{F}$ of $L^V_{\omega_1\omega}$, then $\mathfrak{m}$ is realized if and only if it extends to an $S^{\mathcal{F}}_C$-Henkin type, where $S^{\mathcal{F}}_C$ is the set of sentences of the fragment $\mathcal{F}_C$ of $L^C_{\omega_1\omega}$ generated by $\mathcal{F}$.

*Proof.* The idea is that, since $V$ is countable, a $V$-structure has countable sorts exactly when it has a canonical expansion to $V_C$. By Lemma 2, moreover, the types of canonical $V_C$-structures are exactly the Henkin types. In part (b), Lemma 3 first replaces the realizing structure by one whose sorts are countable. (a) Suppose first that a $V$-structure $\mathfrak{M}$ whose sorts are countable realizes $\mathfrak{m}$. Since every sort of $\mathfrak{M}$ is countable and nonempty, there is an expansion $\mathfrak{M}[C]$ of $\mathfrak{M}$ to $V_C$ in which, for each $M \in \mathbb{M}$, the map $n \mapsto (c_n^M)^{\mathfrak{M}[C]}$ carries $\omega$ onto $M^{\mathfrak{M}}$, and every such expansion is canonical. By Lemma 2(a), applied to $L^C_{\omega_1\omega}$, the full type $S^C_{\omega_1\omega}\text{-tp}(\mathfrak{M}[C])$ is a Henkin type. This Henkin type extends $\mathfrak{m}$, because the value of a $V$-sentence in $\mathfrak{M}[C]$ is its value in $\mathfrak{M}$. Conversely, let $\mathfrak{m}' \in \prod S^C_{\omega_1\omega}$ be a Henkin type that extends $\mathfrak{m}$. By Lemma 2(c), applied to $L^C_{\omega_1\omega}$, the type $\mathfrak{m}'$ is the full type of a canonical $V_C$-structure $\mathfrak{N}$, whose $V$-reduct therefore realizes $\mathfrak{m}$. Since $V_C$ is countable together with $V$, there are only countably many closed $V_C$-terms. The sorts of the $V$-reduct of $\mathfrak{N}$ are therefore countable, every element of $\mathfrak{N}$ being the value of such a term.

(b) If a $V$-structure $\mathfrak{M}$ realizes $\mathfrak{m}$, then so does a $V$-structure $\mathfrak{M}_0$ whose sorts are countable, by Lemma 3 and the inclusion $\Phi \subseteq S^{\mathcal{F}}$. By Lemma 2(a), applied to $\mathcal{F}_C$, the $S_C^{\mathcal{F}}$-type of the canonical expansion $\mathfrak{M}_0[C]$ constructed in (a) is an $S_C^{\mathcal{F}}$-Henkin type, and it extends $\mathfrak{m}$. Conversely, let $\mathfrak{m}'$ be an $S_C^{\mathcal{F}}$-Henkin type that extends $\mathfrak{m}$. By Lemma 2(c), applied to $\mathcal{F}_C$, the type $\mathfrak{m}'$ is the $S_C^{\mathcal{F}}$-type of a canonical $V_C$-structure, whose $V$-reduct realizes $\mathfrak{m}$. □

## 10. Topology on the Space of Henkin Types

The omitting types theorem of Section 11 is obtained from the Baire category theorem, which requires a Čech-complete space. The space in question consists of the Henkin types for the sentences of a fragment of the expanded language, regarded as a subspace of the compact Hausdorff space of all formal types for these sentences. We show that, when the vocabulary and the fragment are countable, the Henkin types form a $G_\delta$ subset of that compact space, and therefore a Čech-complete space and a Baire space.

Let $\mathcal{F}$ be a fragment of $L^V_{\omega_1\omega}$, and recall from Section 6 that $S_C^{\mathcal{F}}$ is the set of sentences of the fragment $\mathcal{F}_C$ of $L^C_{\omega_1\omega}$ generated by $\mathcal{F}$. We denote the set of $S_C^{\mathcal{F}}$-Henkin types by $\mathcal{H}_C^{\mathcal{F}}$. As recalled in Section 2, the product $\prod S_C^{\mathcal{F}} = \prod_{\varphi\in S_C^{\mathcal{F}}}[-r_\varphi, r_\varphi]$ of Section 8, where $r_\varphi$ is the a priori bound of $\varphi$, is a compact Hausdorff space when endowed with the product topology. The product topology is generated by the projections (see [Eng89, Section 2.3 and Proposition 1.4.8]), and the open rays with rational endpoints form a subbasis of the topology of each interval $[-r_\varphi, r_\varphi]$. Together, these two facts show that the sets

$$\langle\!\langle \varphi > r \rangle\!\rangle \coloneqq \left\{ \mathfrak{m} \in \prod S_C^{\mathcal{F}} \colon \varphi^{\mathfrak{m}} > r \right\}$$
$$\langle\!\langle \varphi < r \rangle\!\rangle \coloneqq \left\{ \mathfrak{m} \in \prod S_C^{\mathcal{F}} \colon \varphi^{\mathfrak{m}} < r \right\}$$

where $\varphi \in S_C^{\mathcal{F}}$ and $r \in \mathbb{Q}$, form a subbasis of the product topology. For an $S_C^{\mathcal{F}}$-theory $T$, we write $\langle\!\langle T \rangle\!\rangle$ for the set of formal types $\mathfrak{m} \in \prod S_C^{\mathcal{F}}$ that satisfy every condition of $T$, in the sense of Section 8. We say that $\mathfrak{m}$ **satisfies** $T$ if $\mathfrak{m} \in \langle\!\langle T \rangle\!\rangle$; thus each $\langle\!\langle T \rangle\!\rangle$ is the intersection of the subbasic sets given by the conditions of $T$, and the set $\langle\!\langle \emptyset \rangle\!\rangle$ is all of $\prod S_C^{\mathcal{F}}$. For each $\varphi \in S_C^{\mathcal{F}}$, the projection $\mathfrak{m} \mapsto \varphi^{\mathfrak{m}}$ is continuous on $\prod S_C^{\mathcal{F}}$ (see [Eng89, Proposition 1.4.8]). Hence, for a closed condition $\varphi \geq r$ with $\varphi \in S_C^{\mathcal{F}}$, the set $\langle\!\langle \varphi \geq r \rangle\!\rangle = \{\mathfrak{m} \in \prod S_C^{\mathcal{F}} \colon \varphi^{\mathfrak{m}} \geq r\}$ is closed, being the preimage of the closed ray $[r, \infty)$ under the projection $\mathfrak{m} \mapsto \varphi^{\mathfrak{m}}$. The sets in the proof of Proposition 5 below are likewise built from sets defined by equations and inequalities between finitely many coordinates: if $\sigma_1, \ldots, \sigma_k \in S_C^{\mathcal{F}}$ and $g \colon \mathbb{R}^k \to \mathbb{R}$ is continuous, then the set $\{\mathfrak{m} \colon g(\sigma_1^{\mathfrak{m}}, \ldots, \sigma_k^{\mathfrak{m}}) > 0\}$ is open. Under the same hypotheses, the set $\{\mathfrak{m} \colon g(\sigma_1^{\mathfrak{m}}, \ldots, \sigma_k^{\mathfrak{m}}) = 0\}$, being the intersection of the open sets $\{\mathfrak{m} \colon |g(\sigma_1^{\mathfrak{m}}, \ldots, \sigma_k^{\mathfrak{m}})| < 1/n\}$ with $n \in \mathbb{Z}_+$, is a $G_\delta$ set.

**Proposition 5.** If the vocabulary $V$ and the fragment $\mathcal{F}$ are countable, then the set $\mathcal{H}_C^{\mathcal{F}}$ of $S_C^{\mathcal{F}}$-Henkin types is a $G_\delta$ subset of $\prod S_C^{\mathcal{F}}$.

*Proof.* The idea is that the Henkin conditions consist of countably many equations, each either between finitely many coordinates or between one coordinate and the supremum of countably many others. Each such equation is, moreover, equivalent to countably many requirements defining $G_\delta$ sets. All the index sets below are countable: since $V$ is countable, so are $V_C$, the set of variables and the set of closed $V_C$-terms. The fragment $\mathcal{F}_C$ is countable as well, by Section 6.

By the continuity of the projections, every set in the following list is a $G_\delta$ subset of $\prod S_C^{\mathcal{F}}$. In addition, for each $i \in \{1, \dots, 5\}$, a formal type belongs to all the sets listed under (H$i$) if and only if it satisfies the Henkin condition (H$i$):

(H1) the set $\{\mathfrak{m}\colon 1^{\mathfrak{m}} = 1\}$;

(H2) for $\varphi \in S_C^{\mathcal{F}}$ and $s \in \mathbb{Q}$, the set $\{\mathfrak{m}\colon (s\cdot\varphi)^{\mathfrak{m}} = s\,\varphi^{\mathfrak{m}}\}$;

(H3) for each sentence $\bigvee\Phi \in S_C^{\mathcal{F}}$, the closed sets $\{\mathfrak{m}\colon (\bigvee\Phi)^{\mathfrak{m}} \geq \varphi^{\mathfrak{m}}\}$ with $\varphi \in \Phi$, and the set

$$\bigcap_{n\in\mathbb{Z}_+}\bigcup_{\varphi\in\Phi}\left\{\mathfrak{m}\colon (\bigvee\Phi)^{\mathfrak{m}} < \varphi^{\mathfrak{m}} + \tfrac{1}{n}\right\},$$

which is a $G_\delta$ set because each union is open;

(H4) for each variable $x$ and each formula $\varphi \in \mathcal{F}_C$ whose free variables are among $x$, the closed sets $\{\mathfrak{m}\colon (\sup_x \varphi)^{\mathfrak{m}} \geq \varphi(\tau)^{\mathfrak{m}}\}$ with $\tau$ a closed $V_C$-term of sort ${}_xM$, and the set

$$\bigcap_{n\in\mathbb{Z}_+}\bigcup_{\tau}\left\{\mathfrak{m}\colon (\sup_x \varphi)^{\mathfrak{m}} < \varphi(\tau)^{\mathfrak{m}} + \tfrac{1}{n}\right\};$$

(H5) for $\varphi, \psi \in S_C^{\mathcal{F}}$, the set $\{\mathfrak{m}\colon (\varphi\vee\psi)^{\mathfrak{m}} = \max\{\varphi^{\mathfrak{m}}, \psi^{\mathfrak{m}}\}\}$.

Each set listed under (H1), (H2) and (H5), and each closed set listed under (H3) and (H4), is the zero set of a continuous function of finitely many coordinates (such as $\mathfrak{m} \mapsto \max\{\varphi^{\mathfrak{m}} - (\bigvee\Phi)^{\mathfrak{m}}, 0\}$). Each union listed is a union of sets of the form $\{\mathfrak{m}\colon g(\sigma_1^{\mathfrak{m}}, \dots, \sigma_k^{\mathfrak{m}}) > 0\}$; in both cases, the remark preceding the proposition therefore applies. Being indexed by countable sets, the family of all these sets is countable, and its intersection is $\mathcal{H}_C^{\mathcal{F}}$. Since a countable intersection of $G_\delta$ sets is a $G_\delta$ set, the proposition follows. □

**Corollary 6.** If the vocabulary $V$ and the fragment $\mathcal{F}$ are countable, then $\mathcal{H}_C^{\mathcal{F}}$ is Čech-complete.

*Proof.* By Proposition 5, the space $\mathcal{H}_C^{\mathcal{F}}$ is a $G_\delta$ subset of the compact Hausdorff space $\prod S_C^{\mathcal{F}}$; as recalled in Section 2, every such subset is Čech-complete. □

**Corollary 7.** If the vocabulary $V$ and the fragment $\mathcal{F}$ are countable, then $\mathcal{H}_C^{\mathcal{F}}$ is a Baire space.

*Proof.* By Corollary 6, the space $\mathcal{H}_C^{\mathcal{F}}$ is Čech-complete, and the Baire category theorem for Čech-complete spaces (see Theorem 1) shows that it is a Baire space. □

In Section 11 we express in terms of conditions that a set of Henkin types has empty interior. For this purpose we use a basis of the topology of $\mathcal{H}_C^{\mathcal{F}}$ described by

finite theories. Since $\langle\!\langle S \rangle\!\rangle$ is a finite intersection of subbasic open sets whenever $S$ is a finite $S^{\mathcal{F}}_C$-theory, the sets $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle S \rangle\!\rangle$ with $S$ finite form a basis of the topology of $\mathcal{H}^{\mathcal{F}}_C$ (see [Eng89, Sections 1.1 and 2.1]).

## 11. Omitting Types Theorem

The Henkin types satisfying a theory are the types of the canonical structures (those in which every element is the value of a closed term) whose reducts are models of the theory. To omit countably many types in a model of the theory, it therefore suffices to find such a Henkin type that realizes none of them. In this section we show that, when the vocabulary and the fragment are countable and the theory has a model, the Henkin types satisfying the theory form a nonempty Baire space. We also show that the required Henkin type exists if, for each instance of each of the given types, the members of this space satisfying the instance form a nowhere dense set. The members of this space that realize one of the given types then form a meager subset, and the Baire category theorem provides a Henkin type outside it.

Henceforth, assume that the vocabulary $V$ and the fragment $\mathcal{F}$ of Section 10 are countable. We also fix an $S^{\mathcal{F}}$-theory $B$ (a theory whose conditions are on sentences of $\mathcal{F}$) that has a model. Recall from Section 10 that $\mathcal{H}^{\mathcal{F}}_C$ is the set of $S^{\mathcal{F}}_C$-Henkin types, where $S^{\mathcal{F}}_C$ is the set of sentences of the fragment $\mathcal{F}_C$ of $L^C_{\omega_1\omega}$ generated by $\mathcal{F}$. The set $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$ of $S^{\mathcal{F}}_C$-Henkin types that satisfy $B$ (see Section 10) is nonempty. Indeed, by Lemma 3, the theory $B$ has a model $\mathfrak{M}$ whose sorts are countable. The canonical expansion $\mathfrak{M}[C]$ of $\mathfrak{M}$ constructed in the proof of Proposition 4(a) satisfies $B$, since the values of $V$-sentences in $\mathfrak{M}[C]$ are their values in $\mathfrak{M}$. By Lemma 2(a), the $S^{\mathcal{F}}_C$-type of $\mathfrak{M}[C]$ belongs to $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$. Conversely, by Lemma 2(c), every member of $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$ is the $S^{\mathcal{F}}_C$-type of a canonical $V_C$-structure, whose $V$-reduct is a model of $B$. Thus the members of $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$ are exactly the $S^{\mathcal{F}}_C$-types of the canonical $V_C$-structures whose $V$-reducts are models of $B$. Since $S^{\mathcal{F}}$ is countable, so is the theory $B$, whence the set $\langle\!\langle B \rangle\!\rangle$ is a countable intersection of open sets. By Proposition 5, the set $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$ is therefore a $G_\delta$ subset of the compact Hausdorff space $\prod S^{\mathcal{F}}_C$. As recalled in Section 2, a $G_\delta$ subset of a compact Hausdorff space is Čech-complete; by Theorem 1, the space $\mathcal{H}^{\mathcal{F}}_C \cap \langle\!\langle B \rangle\!\rangle$ is a nonempty Baire space.

Let $\bar{x}$ be a finite tuple of distinct variables. An **$\mathcal{F}$-type in** $\bar{x}$ is a set $T(\bar{x})$ of expressions $\varphi(\bar{x}) \geq r$, where $r \in \mathbb{Q}$ and $\varphi \in \mathcal{F}$ is a formula whose free variables are among $\bar{x}$. Substituting closed terms for $\bar{x}$ turns each of these expressions into a closed condition in the sense of Section 7. For a tuple $\bar{\tau}$ of closed $V_C$-terms of sort ${}_{\bar{x}}M$, the **instance** $T(\bar{\tau})$ of $T(\bar{x})$ is the union of the closed conditions $\varphi(\bar{\tau}) \geq r$ for the members $\varphi(\bar{x}) \geq r$ of $T(\bar{x})$. Each of these conditions is a countable theory; the instance $T(\bar{\tau})$ is an $S^{\mathcal{F}}_C$-theory because the fragment $\mathcal{F}_C$ is closed under substitution of terms (see (F4) in Section 6). An $S^{\mathcal{F}}_C$-Henkin type $\mathfrak{m} \in \mathcal{H}^{\mathcal{F}}_C$ **realizes** $T(\bar{x})$ if it satisfies some instance $T(\bar{\tau})$, that is, if $\mathfrak{m} \in \langle\!\langle T(\bar{\tau}) \rangle\!\rangle$ for some tuple $\bar{\tau}$; otherwise it **omits** $T(\bar{x})$. Similarly, if $\mathfrak{M}$ is a $V$-structure, a tuple $\bar{a} \in {}_{\bar{x}}M^{\mathfrak{M}}$ **satisfies** $T(\bar{x})$ in $\mathfrak{M}$ if $\varphi^{\mathfrak{M}}(\bar{a}) \geq r$ for every member $\varphi(\bar{x}) \geq r$ of $T(\bar{x})$. We say that $T(\bar{x})$ **has**

**nonempty interior in** $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ if there exist a tuple $\bar{\tau}$ of closed $V_C$-terms of sort ${}_{\bar{x}}M$ and a finite $S_C^{\mathcal{F}}$-theory $S$ such that

$$\emptyset \neq \mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle \cap \langle\langle S \rangle\rangle \subseteq \langle\langle T(\bar{\tau}) \rangle\rangle .$$

Otherwise, we say that $T(\bar{x})$ **has empty interior in** $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. The sets $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle \cap \langle\langle S \rangle\rangle$, with $S$ a finite $S_C^{\mathcal{F}}$-theory, form a basis of the topology of $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ (see Section 10). Consequently, the type $T(\bar{x})$ has nonempty interior in this space exactly when, for some tuple $\bar{\tau}$, the closed subset $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle \cap \langle\langle T(\bar{\tau}) \rangle\rangle$ of this space has nonempty interior. An $\mathcal{F}$-type is expressed by closed conditions so that this subset, being an intersection of the closed sets $\langle\langle \varphi(\bar{\tau}) \geq r \rangle\rangle$, is closed (see Section 10). When this subset has empty interior, its complement in $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ is therefore a dense open set. The Baire category theorem applies to countable intersections of such dense open sets, as the proof of Theorem 8 shows.

**Theorem 8** (Omitting Types Theorem)**.** Assume that the vocabulary $V$ and the fragment $\mathcal{F}$ are countable, and let $B$ be an $S^{\mathcal{F}}$-theory that has a model. Let $(T_n(\bar{x}_n))_{n<\omega}$ be a sequence of $\mathcal{F}$-types each of which has empty interior in $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. Then there is an $S_C^{\mathcal{F}}$-Henkin type $\mathfrak{m} \in \mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ that omits every $T_n(\bar{x}_n)$. Consequently, the theory $B$ has a model whose sorts are countable and in which, for every $n < \omega$, no tuple of elements satisfies $T_n(\bar{x}_n)$.

*Proof.* The idea is to write the set of Henkin types that satisfy $B$ and omit every $T_n(\bar{x}_n)$ as a countable intersection of open dense subsets of the nonempty Baire space $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. Fix $n < \omega$, let $\bar{x} \coloneqq \bar{x}_n$ and $T \coloneqq T_n$, and let $\bar{\tau}$ be a tuple of closed $V_C$-terms of sort ${}_{\bar{x}}M$. Since the set $\langle\langle \varphi(\bar{\tau}) \geq r \rangle\rangle$ is closed for every member $\varphi(\bar{x}) \geq r$ of $T(\bar{x})$ (see Section 10), the set $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle \cap \langle\langle T(\bar{\tau}) \rangle\rangle$ is closed in the space $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. By the definition of empty interior, this closed set has empty interior in $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$, and the complement

$$O_{T,\bar{\tau}} \coloneqq \{\mathfrak{m} \in \mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle : \mathfrak{m} \text{ does not satisfy } T(\bar{\tau})\}$$

is therefore an open dense subset of $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. Since $V$ is countable, there are only countably many tuples $\bar{\tau}$; consequently, the set $O_T \coloneqq \bigcap_{\bar{\tau}} O_{T,\bar{\tau}}$ of members of $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ that omit $T(\bar{x})$ is a countable intersection of open dense subsets of $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$. As $n$ was arbitrary, the same holds for each set $O_{T_n}$, and hence for the set $O \coloneqq \bigcap_{n<\omega} O_{T_n}$ of members of $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ that omit $T_n(\bar{x}_n)$ for every $n < \omega$. Since $\mathcal{H}_C^{\mathcal{F}} \cap \langle\langle B \rangle\rangle$ is a nonempty Baire space, the set $O$ is dense in it, and in particular nonempty.

Let $\mathfrak{m}$ be a member of $O$. By Lemma 2(c), the type $\mathfrak{m}$ is the $S_C^{\mathcal{F}}$-type of a canonical $V_C$-structure $\mathfrak{N}$, whose $V$-reduct $\mathfrak{M}$ is a model of $B$. The sorts of $\mathfrak{M}$ are countable, since every element of $\mathfrak{N}$ is the value of one of the countably many closed $V_C$-terms. Now fix $n < \omega$ and a tuple $\bar{a} \in {}_{\bar{x}_n}M^{\mathfrak{M}}$. Since $\mathfrak{N}$ is canonical, we may write $\bar{a} = \bar{\tau}^{\mathfrak{N}}$ for a tuple $\bar{\tau}$ of closed $V_C$-terms. The substitution lemma (see the proof of Lemma 2(a)), which holds for tuples of closed terms by the same induction, gives $\varphi^{\mathfrak{M}}(\bar{a}) = \varphi^{\mathfrak{N}}(\bar{\tau}^{\mathfrak{N}}) = \varphi(\bar{\tau})^{\mathfrak{m}}$ for every formula $\varphi \in \mathcal{F}$ whose free variables are among $\bar{x}_n$. Since $\mathfrak{m}$ omits $T_n(\bar{x}_n)$, it does not satisfy $T_n(\bar{\tau})$, and the tuple $\bar{a}$ therefore does not satisfy $T_n(\bar{x}_n)$ in $\mathfrak{M}$. □

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